

Exploring the BRFSS data

Setup

Load packages

```
library(ggplot2)
library(dplyr)
```

Load data

Make sure your data and R Markdown files are in the same directory. When loaded your data file will be called `brfss2013`. Delete this note when before you submit your work.

```
target_file <- "alabama_male.RData"

if (file.exists(target_file)) {
  load(target_file)
  df <- df
} else {
  load("brfss2013.RData")

  df <- brfss2013 %>%
    filter(stateres == "Yes", sex == "Male", X_state == "Alabama") %>%
    mutate(
      exercise = ifelse(exerany2 == "Yes", "Active", "Inactive"),
      cholesterol = ifelse(toldhi2 == "Yes", "High", "Normal"),
      drinks = ifelse(X_drnkmo4 <= 9998, X_drnkmo4, NA)) %>%
    filter(!is.na(exercise), !is.na(cholesterol)) %>%
    select(exercise, cholesterol, drinks)

  save(df, file = target_file)
  rm(brfss2013)
}
```

Part 1: Data

The Behavioral Risk Factor Surveillance System (BRFSS) is a collaborative project between U.S. states and the Centers for Disease Control and Prevention (CDC). As an ongoing surveillance system, it measures behavioral risk factors for the non-institutionalized adult population (18 years and older) residing in the United States.

Data Collection Method

The data are collected through monthly telephone interviews using a stratified multistage sampling design. Sampling is first stratified by state, followed by a multistage cluster sampling approach within each state to select households and individuals randomly. This sampling method ensures proportionate representation across geographic and demographic segments.

Scope of Inference

Generalizability: The observations are collected through a random-selection process within a multistage sampling framework, the results are highly representative. Thus, findings can be generalized to the non-institutionalized adult population of the U.S., specifically to Alabama for this study. **Causality:** This is an observational study based on cross-sectional surveys rather than a controlled experiment. Therefore, we can only establish associations(correlations) as there is no random assignment to treatment groups. (e.g., participants were not randomly assigned to drink or exercise.)

Purpose of the Study

This study investigates how lifestyle choices specifically physical activity and alcohol consumption relate to cholesterol levels among Alabama men. Understanding these associations is critical for developing effective public health interventions against cardiovascular risk.

Variables

- **Exercise (exercise):** Nominal categorical (Active / Inactive).
 - **Cholesterol (cholesterol):** Nominal categorical (High / Normal).
 - **Drink habits (drinks):** Continuous numerical (Monthly alcohol intake).
 - **State, Sex, Stateres:** These variables are used as filtering criteria to define the study population (i.e., male residents in Alabama). They ensure that demographic factors are held constant across the analysis.
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Part 2: Research questions

Research question 1: Is there a significant association between exercise status (Active vs. Inactive) and high cholesterol diagnosis among men in Alabama?

- **Interest:** Physical inactivity is a well-known risk factor for cardiovascular disease. By analyzing this relationship, we can determine how much a simple lifestyle change - regular exercise-is associated with better cholesterol management in this specific demographic.

Research question 2: Is the association between alcohol consumption and high cholesterol different depending on a person's exercise status (Active vs. Inactive) among Alabama men?

Interest: This question is interesting because it examines whether regular exercise can mitigate the potential negative effects of alcohol on cholesterol levels. It involves three variables (drinks, exercise, and cholesterol) to see if they interact.

Research question 3: How does the presence of heavy drinkers (outliers in alcohol consumption) affect the observed relationship between exercise and high cholesterol?

- Interest: Statistical averages can be misleading if skewed by outliers. This question investigates whether the health benefits of exercise remain consistent across different levels of alcohol consumption, or if extreme drinking habits distort the overall correlation between lifestyle and cholesterol. (Variables: drinks, exercise, cholesterol)
-

Part 3: Exploratory data analysis

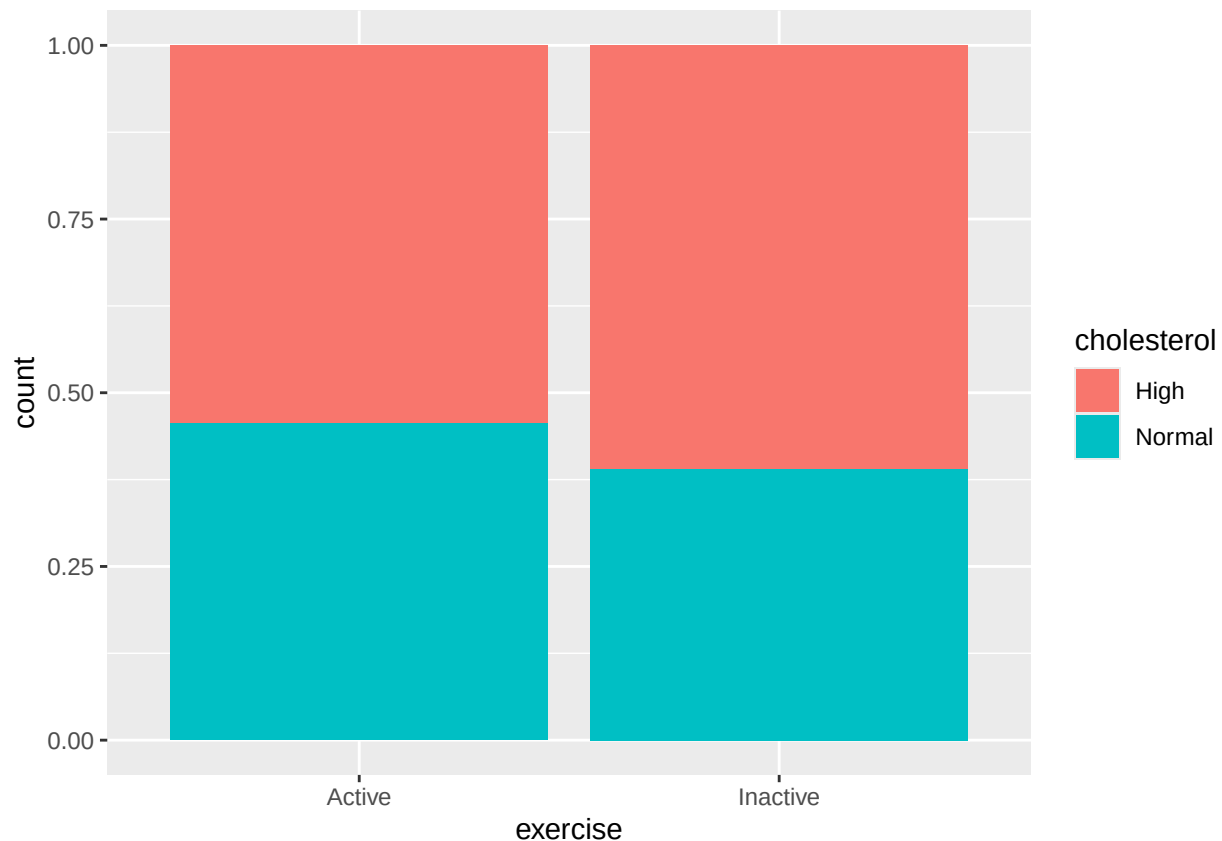
```
df <- brfss2013 %>%
filter(stateres == "Yes", sex == "Male", X_state=="Alabama") %>%
  mutate(
    exercise = ifelse(exerany2 == "Yes", "Active", "Inactive"),
    cholesterol = ifelse(toldhi2 == "Yes", "High", "Normal"),
    drinks = ifelse(X_drnkmo4 <= 9998, X_drnkmo4, NA)) %>%
filter(!is.na(exercise), !is.na(cholesterol)) %>%
select(X_state, exercise, cholesterol, drinks)

save(df, file = "alabama_male.RData")
rm(brfss2013)
```

Research question 1: Is there a significant association between exercise status (Active vs. Inactive) and high cholesterol diagnosis among men in Alabama?

```
df$exercise <- as.factor(df$exercise)
df$cholesterol <- as.factor(df$cholesterol)

df %>%
  ggplot(aes(x = exercise, fill = cholesterol)) +
  geom_bar(position = "fill")
```



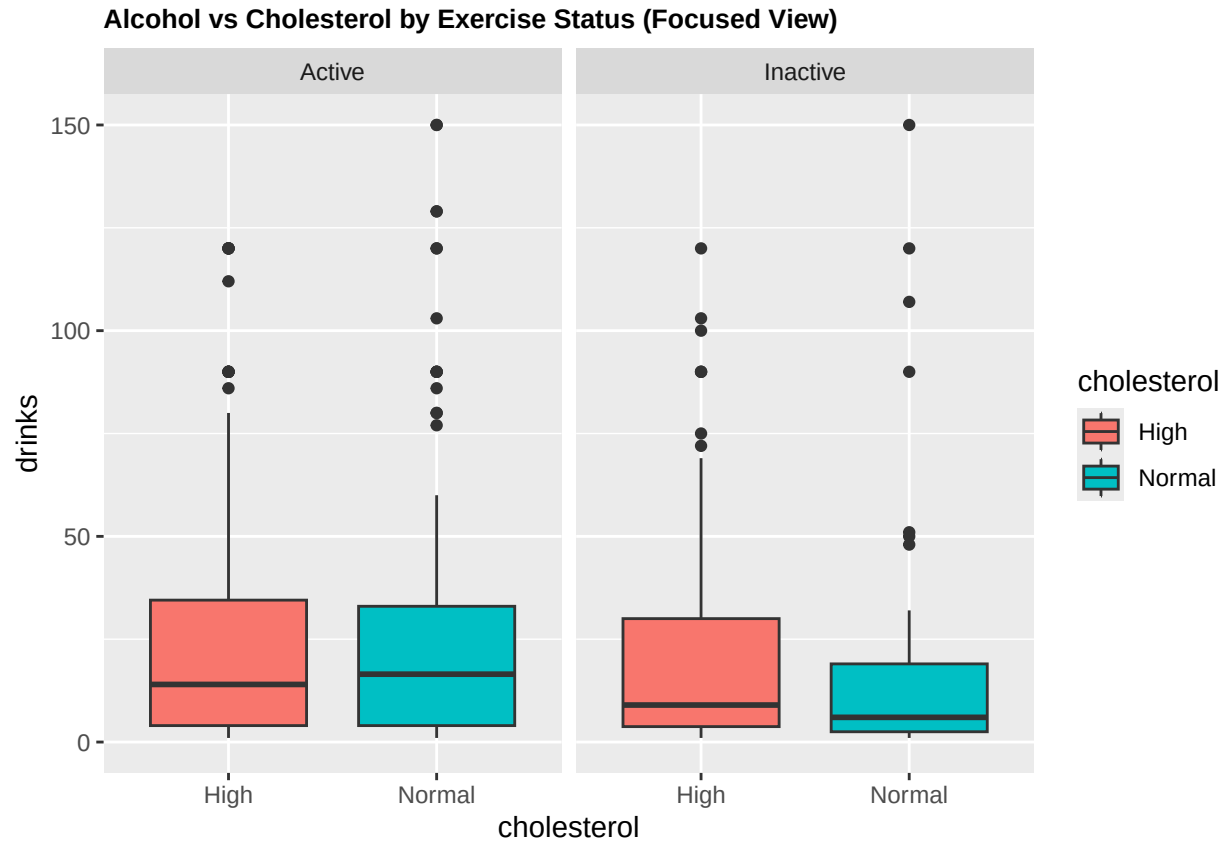
Analysis to question 1:

The stacked bar chart illustrates the association between exercise status and cholesterol levels. The inactive group shows higher proportion of high cholesterol compared to the active group.

- **Conclusion:** There is a noticeable association between exercise status and cholesterol levels, where a lack of regular exercise is linked to a higher prevalence of high cholesterol.

Research question 2: Is the association between alcohol consumption and high cholesterol different depending on a person's exercise status (Active vs. Inactive) among Alabama men who drink at least once a month?

```
df %>%
  filter(drinks>0) %>%
  ggplot(aes(x=cholesterol, y= drinks, fill = cholesterol))+
  geom_boxplot()+
  facet_wrap(~ exercise) +
  coord_cartesian(ylim = c(0,150)) +
  labs(title = "Alcohol vs Cholesterol by Exercise Status (Focused View)") +
  theme(plot.title = element_text(size=10, face="bold"))
```



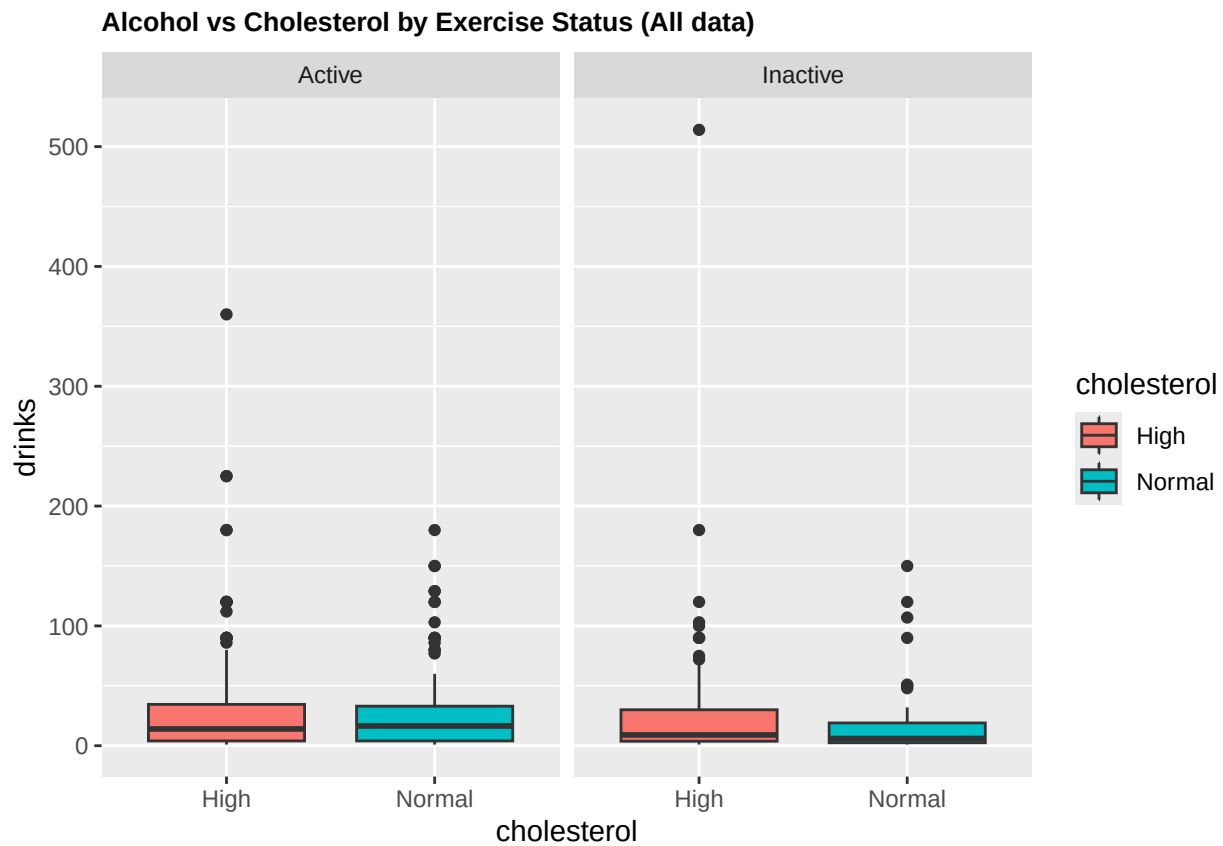
Analysis to question 2:

The box plots above show the association between drink consumption and cholesterol in two groups who exercise and who do not.

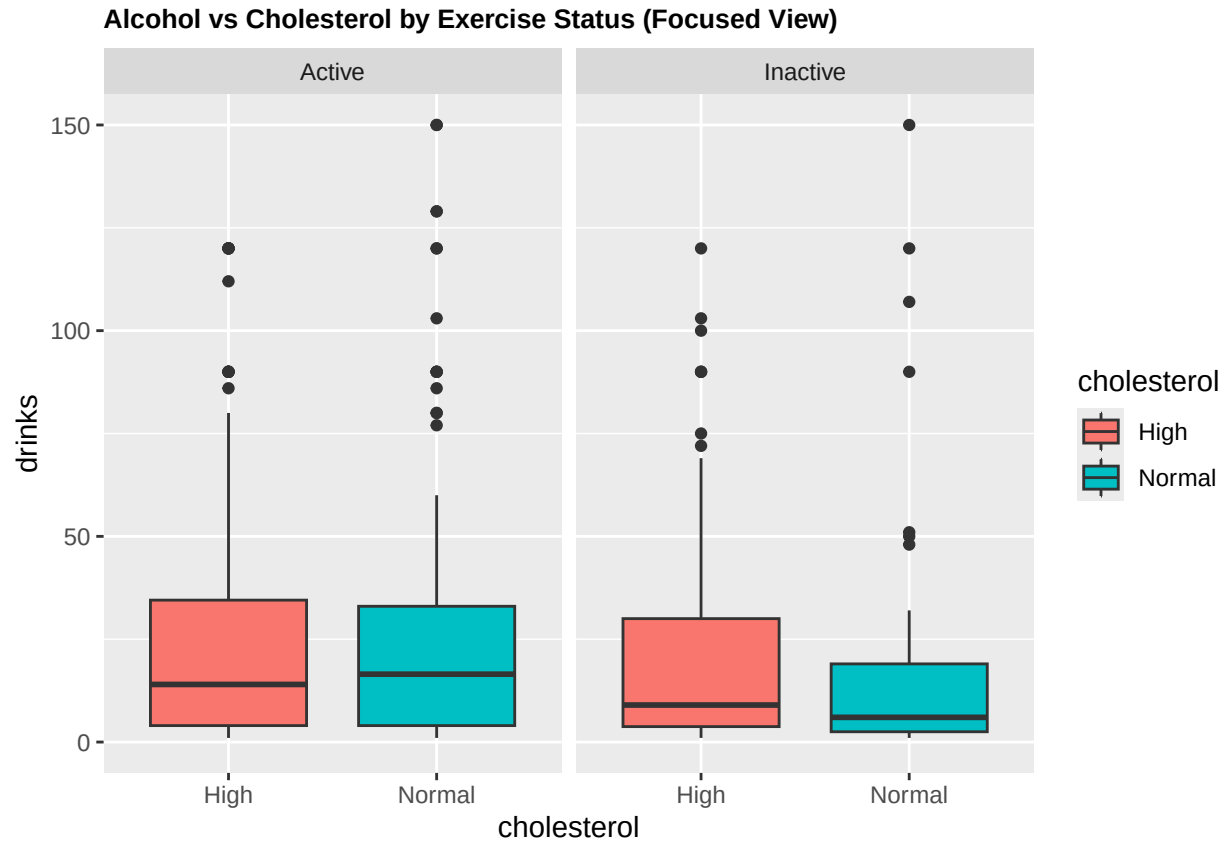
- **Exercise Group(Active):** There is no clear association between high cholesterol diagnosis and monthly drink intake. Interestingly, the median drink intake is slightly higher in the normal cholesterol group. The wider Interquartile Range (IQR) in this group suggests a more diverse drinking pattern among active individuals, yet this does not directly translate to a higher prevalence of high cholesterol.
- **Non-Exercise Group(Inactive):** In contrast, the normal cholesterol group exhibits the most condensed distribution and the lowest median alcohol consumption among all four sub-groups. This indicates that those who do not exercise, lower alcohol intake is more strongly associated with maintaining normal cholesterol levels.
- **Note on Outliers:** While the actual dataset includes significant outliers (heavy drinkers), this analysis focuses on the range up to 150 monthly drinks. This allows for a more accurate evaluation of the median and typical drinking patterns across the groups without the scale being distorted by extreme values.
- **Conclusion:** Exercise status appears to be a significant variable that moderates the relationship between drink consumption and cholesterol levels. The data suggests that the association between drinking and cholesterol is more evident in the absence of regular exercise.

Research question 3: How does the presence of heavy drinkers (outliers in alcohol consumption) affect the observed relationship between exercise and high cholesterol?

```
df %>%
  filter(drinks>0) %>%
  ggplot(aes(x=cholesterol, y= drinks, fill = cholesterol))+
  geom_boxplot()+
  facet_wrap(~ exercise) +
  coord_cartesian(ylim = c(0,515)) +
  labs(title = "Alcohol vs Cholesterol by Exercise Status (All data)")+
  theme(plot.title = element_text(size=10, face="bold"))
```



```
df %>%
  filter(drinks>0) %>%
  ggplot(aes(x=cholesterol, y= drinks, fill = cholesterol))+
  geom_boxplot()+
  facet_wrap(~ exercise) +
  coord_cartesian(ylim = c(0,150)) +
  labs(title = "Alcohol vs Cholesterol by Exercise Status (Focused View)") +
  theme(plot.title = element_text(size=10, face="bold"))
```



Analysis to question 3:

This visualization with the full data range (up to 515 drinks per month) reveals that extreme values are present in both exercise groups, particularly within the high cholesterol categories.

- **Impact on Distribution:** These outliers contribute significantly to the right-skewness of the data, which underscores the importance of using medians rather than means for a representative comparison.
- **Observed Association:** Interestingly, these heavy drinkers are frequently found in both high cholesterol groups regardless of their exercise status.
- **Conclusion:** This suggests a notable association between the presence of heavy drinkers and a high cholesterol diagnosis. Therefore, while outliers can distort the overall scale, they also provide critical insight into the extreme drinking behaviors linked to cardiovascular risk factors.

Conclusion of the Study

This study reveals that the majority men in Alabama do not consume alcohol regularly. However, when the scope is narrowed to those who drink at least once a month, two notable findings emerge:

- **Non-Exercise Group(Inactive):** Among men who do not exercise regularly, there is a clear association between lower alcohol consumption and normal cholesterol levels.
- **Exercise Group(Active):** For men who exercise regularly, there is no strong evidence of a direct relationship between alcohol intake and cholesterol status. Instead, the normal cholesterol group within

this category exhibits more diverse drinking habits compared to their inactive counterparts, suggesting that exercise may serve as a protective or moderating factor.

- **Conclusion:** Therefore, public health interventions aimed at reducing cardiovascular risk should be strategically tailored. For sedentary individuals, coaching to reduce alcohol consumption is likely to be highly effective. Overall, promoting a combination of regular exercise and moderate drinking remains a key strategy for managing cholesterol-related health risks in population.